NAMED ENTITY RECOGNITION FOR THE DISAMBIGUATION OF SPECIFIC NUMERALS IN ROMANIAN

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INTRODUCTION

Research Background

- **Context**: Specialized extension to a robust hybrid system for numeral transcription.
- **Architecture**: AI-based models (NER, BERT) combined with rule-based deterministic filters
- **Significance**: Accurate numeral transcription is crucial for data integrity.
- **Objective**: Enhance semantic precision by addressing specific edge cases in Romanian language

Problem Statement

The "Un / O" Ambiguity

- Persistent ambiguity between indefinite articles and numerals quantifying humans.
- Generic NER models struggle with contextual disambiguation of these forms
- The goal is to explicitly label human-centric numerals using a fine-tuned Romanian BERT model.

Examples

- "**un** accident" (= an accident): indefinite article, ignored by the module
- "**un** barbat" (= a man): human numeral, target for quantification

Disambiguation Pipeline

INPUT

Raw Romanian text containing mixed numeral forms and structured identifiers.

MASKING

Regex-driven identification and protection of standardized numeric codes (e.g., IBAN, VIN).

DISAMBIGUATION (Our model)

Fine-tuned transformer classification of "un" and "o" within human-centric semantic contexts.

CONVERSION

Rule-based reconstruction by turning validated spans into digits.

- **Strategic Integration**: Acting as a granular filter before the conversion stage.
- **Layered Architecture**
- **Efficiency**: Maintains system speed by using BERT inference strictly for ambiguous spans.

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THE DATASET & ANNOTATION STRATEGY

Dataset & BIO Tagging

Refined Training Set

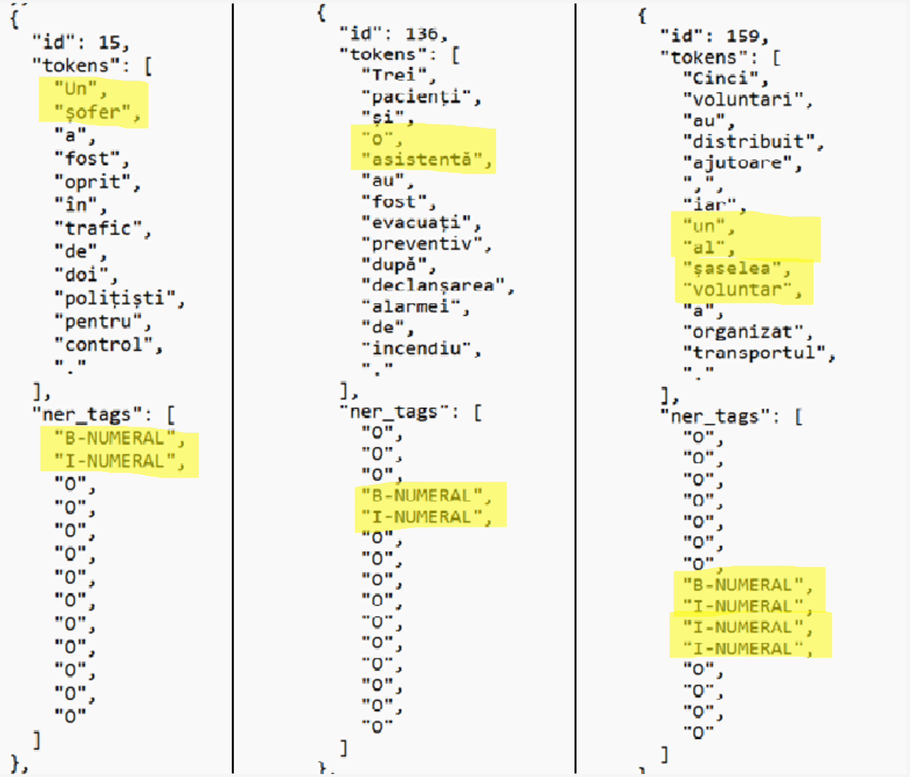
Manually annotated Romanian text specifically for human-quantifying numerals, stored in JSON format.

BIO Scheme

Utilizing the **Beginning-Inside-Outside** tagging strategy for precise delimitation.

Contextual diversity

Incorporating administrative, legal and daily-life text for cross-domain robustness.



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    "de",
    "doi",
    "polițiști",
    "pentru",
    "control",
    "."
  ],
  "ner_tags": [
    "B-NUMERAL",
    "I-NUMERAL",
    "O",
    "O",
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    "O",
    "O",
    "O",
    "O",
    "O",
    "O",
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},
```

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    "și",
    "o",
    "asistentă",
    "au",
    "fost",
    "evacuați",
    "preventiv",
    "după",
    "declanșarea",
    "alarmei",
    "de",
    "incendiu",
    "."
  ],
  "ner_tags": [
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    "O",
    "O",
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    "I-NUMERAL",
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    "O",
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    "O",
    "O"
  ]
},
```

```
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    "au",
    "distribuit",
    "ajutoare",
    ",",
    "iar",
    "un",
    "al",
    "șaselea",
    "voluntar",
    "a",
    "organizat",
    "transportul",
    "."
  ],
  "ner_tags": [
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    "O",
    "O",
    "O",
    "O",
    "O",
    "O",
    "B-NUMERAL",
    "I-NUMERAL",
    "I-NUMERAL",
    "I-NUMERAL",
    "O",
    "O",
    "O",
    "O"
  ]
}
```


Examples of annotated phrases

Linguistic Traps

- **Beyond Simple Cases**: Moving from standard **[numeral + noun]** pairs to complex, deceptive structures

- **Hard Negatives**:

➢ <u>Interjections</u>: “**O**, ce bine!” (= **Oh**, how nice!) => identified as *OUTSIDE*

➢ <u>Personified Objects</u>: “**Un** robot utilitar [...]” (=A utility robot) => filtered out

- **Complex Positives**: “**Un** purtător de cuvânt [...]” (= A spokesperson)

03

METHODOLOGY & TRAINING

Model Architecture

Base Model

Fine-tuned **bert-base-Romanian-cased-v1** transformer.

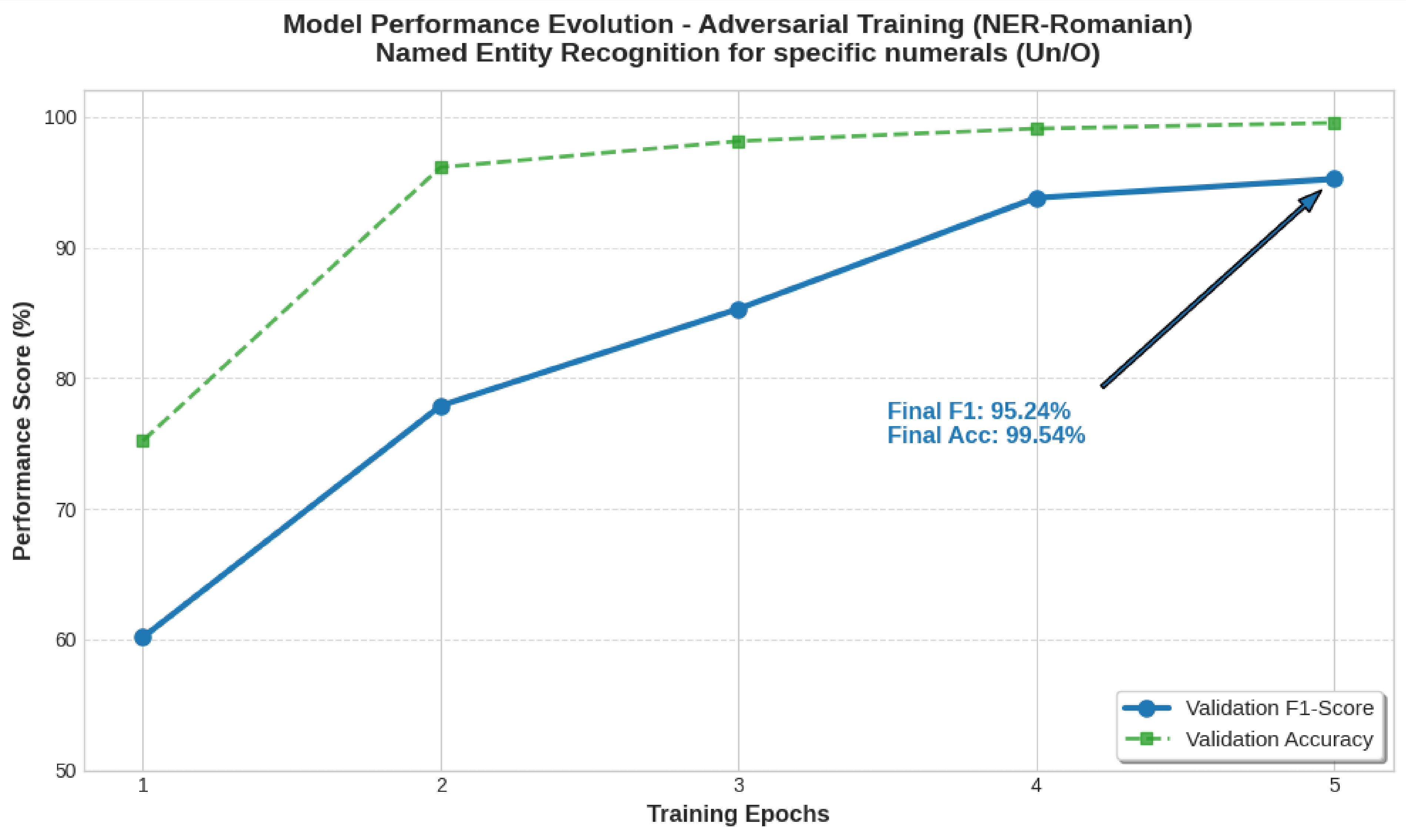
Linguistic Pre-processing

Specialized diacritic normalization (e.g., Ș/Ş, Ț/Ţ) and hyphen standardization.

Input Features

Word-piece tokenization optimized for Romanian morphology.

Training Dynamics



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RESULTS

Practical implementation

```python
def test_model(text):
    results = nlp(text)
    print(f"\nText: {text}")
    if not results:
        print("No specialized numerals detected.")
    for ent in results:
        print(f"-> Detected Entity: '{ent['word']}' | Label: {ent['entity_group']} | Score: {ent['score']:.4f}")

# TEST CASES
test_model("Un tânăr a fost reținut după ce trei persoane au fost agresate într-un bar.")
test_model("Doi cercetători au publicat studiul inițial, iar un al treilea cercetător a contribuit ulterior la analiză.")
test_model("Patru muncitori și un inginer au fost răniți după prăbușirea unei schele pe șantier. Patru programatori au dezvoltat aplicația, iar un al cincilea programator a verificat codul.")
test_model("Am asistat la un accident rutier produs într-un parc aglomerat din centrul vechi al capitalei.")
```

```
Text: Un tânăr a fost reținut după ce trei persoane au fost agresate într-un bar.
-> Detected Entity: 'Un tânăr' | Label: NUMERAL | Score: 0.9983

Text: Doi cercetători au publicat studiul inițial, iar un al treilea cercetător a contribuit ulterior la analiză.
-> Detected Entity: 'un al treilea cercetător' | Label: NUMERAL | Score: 0.9986

Text: Patru muncitori și un inginer au fost răniți după prăbușirea unei schele pe șantier. Patru programatori au dezvoltat aplicația, iar un al cincilea programator a verificat codul.
-> Detected Entity: 'un inginer' | Label: NUMERAL | Score: 0.9964
-> Detected Entity: 'un al cincilea programator' | Label: NUMERAL | Score: 0.9974

Text: Am asistat la un accident rutier produs într-un parc aglomerat din centrul vechi al capitalei.
No specialized numerals detected.
```

Success cases

- **Complex Noun Phrases**: Correct delimitation of phrases like "**un** fost secretar de stat" (= a former Secretary of State).
- **Professional Contexts**: Accurate identification of phrases like "**un** expert in securitate" (= a security expert).
- **Negative Filtering**: Successfully ignored interjections like "**O**, ce veste!" (= **Oh**, what great news!).

Error analysis

- **Identifying Weaknesses**: Initial models struggled with "I-tag" (INSIDE) drop-offs in long noun phrases like "**Un** purtător de cuvânt al Guvernului" (= A government spokesperson), where only "Un purtător" was identified.

- **Performance Jump**: Targeted training raised the F1-Score from 73.24% to **95.24%**.

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CONCLUSIONS

Conclusions & System impact

- **Critical Refinement**: Successfully addressed complex linguistic ambiguities unique to the Romanian language.
- **Data Integrity**: Ensures precise person-centric quantification in sensitive administrative contexts.
- **Performance Validation**: Achieved a **95.24% F1-Score** on datasets containing linguistic traps.

Acknowledgements & Bibliography

Acknowledgements:

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Bibliography:

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- [2] Stefan Dumitrescu, Andrei-Marius Avram, and Sampo Pyysalo. “The birth of Romanian BERT” in *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 4324–4328 [Online]. Association for Computational Linguistics. Available: https://aclanthology.org/2020.findings-emnlp.387
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